STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-DG-HULL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3512-3520 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3522-3552, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_sdga.json (pb=1098, stmt 3512-3520, proof 3522-3552).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AHQ: sources/stacks-sdga/pinned-original-source.tex, source lines 3552–3552. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRJ: sources/stacks-sdga/pinned-original-source.tex, source lines 1444–1455. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. The forgetful functor $F : Mod(\mathcal{A}, d) \to Mod(\mathcal{A})$ has a left adjoint $G : Mod(\mathcal{A}) \to Mod(\mathcal{A}, d)$.*

Proof. To prove the existence of G we can use the adjoint functor theorem, see Categories, Theorem 0AHQ (observe that we have switched the roles of F and G). The exactness conditions on F are satisfied by Lemma 0FRJ. The set theoretic condition can be seen as follows: suppose given a graded $\mathcal{A}$-module $\mathcal{N}$. Then for any map

$$\varphi : \mathcal{N} \longrightarrow F(\mathcal{M})$$

we can consider the smallest differential graded $\mathcal{A}$-submodule $\mathcal{M}' \subset \mathcal{M}$ with $\mathrm{Im}(\varphi) \subset F(\mathcal{M}')$. It is clear that $\mathcal{M}'$ is the image of the map of graded $\mathcal{A}$-modules

$$\mathcal{N} \oplus \mathcal{N}[-1] \otimes_{\mathcal{O}} \mathcal{A} \longrightarrow \mathcal{M}$$

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defined by

$$(n, \sum n_i \otimes a_i) \longmapsto \varphi(n) + \sum \mathrm{d}(\varphi(n_i))a_i$$

because the image of this map is easily seen to be a differential graded submodule of $\mathcal{M}$. Thus the number of possible isomorphism classes of these $\mathcal{M}'$ is bounded and we conclude. $\square$